

Environmental Determinants of Health and Healthcare Accessibility in Rural Border Villages of Jammu in India

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Abstract

Epidemiological patterns in rural India are closely intertwined with environmental and infrastructural determinants, particularly in regions where limited healthcare access and sanitation challenges persist. This study investigates environmental factors influencing public health outcomes in two semi-rural border villages Suchetgarh and Kool Kalan located in the R.S. Pura sub-district of Jammu, India. Primary data were collected from 120 households through structured surveys, complemented by field observations and spatial analysis. Descriptive and spatial methods were employed to examine linkages between sanitation, waste management, water quality, and disease prevalence. Results reveal that poor drainage, open waste disposal, and seasonal waterlogging during monsoon months are major contributors to recurring vector-borne and waterborne illnesses. While Kool Kalan exhibits marginally better access to healthcare and awareness, both villages display environmentally driven disease trends. The findings emphasize the need to integrate environmental management with healthcare delivery to address rural health disparities. This field-based assessment provides empirical evidence on the interaction between environmental determinants and epidemiological vulnerability, supporting localized strategies for sustainable rural health planning aligned with Sustainable Development Goals 3 (Good Health and Well-being) and 6 (Clean Water and Sanitation).

Keywords: Environmental health · Rural healthcare · Epidemiology · Jammu · Sanitation · Sustainable Development Goals

1. Introduction

Epidemiology the study of the distribution and determinants of disease in populations offers vital insights into how biological, environmental, and social factors collectively shape public health outcomes. In recent decades, the field has expanded beyond biomedical determinants to

incorporate environmental and infrastructural contexts, acknowledging that disease prevalence often arises from broader socio-ecological systems (Park, 2021; Prüss-Ustün et al., 2019). This interdisciplinary perspective has given rise to environmental epidemiology, which explores how air and water quality, sanitation, housing, and local ecology influence disease patterns and human well-being (WHO, 2022).

Globally, about one in four deaths is associated with modifiable environmental risks such as polluted air, unsafe water, inadequate sanitation, and exposure to toxic materials (Patel et al., 2022). Rapid urbanization, deforestation, and climate variability have further intensified these risks, particularly in developing countries where basic infrastructure remains inadequate. Environmental degradation and climate-sensitive diseases such as malaria, dengue, and diarrhoeal infections continue to strain public health systems in low- and middle-income regions (Nguyen et al., 2023).

In India, the intersection of environmental conditions and health outcomes is especially complex. Rural and peri-urban areas home to nearly 65 % of the population struggle with sanitation deficits, waste mismanagement, unsafe water sources, and uneven healthcare distribution (NITI Aayog, 2022; Chakraborty & Singh, 2020). These systemic challenges not only expose populations to communicable and non-communicable diseases but also perpetuate socio-environmental inequities. Limited health literacy and inadequate institutional outreach compound these issues, resulting in preventable health burdens (Verma & Joshi, 2023).

The R.S. Pura sub-district of Jammu, located near the international border, exemplifies these conditions. It is a predominantly agrarian region where environmental hazards, infrastructural gaps, and healthcare inaccessibility converge. Within this context, the present study examines two representative villages Suchetgarh and Kool Kalan to analyse the interrelationship between environmental determinants and health outcomes. The objectives are to assess local disease prevalence, public health awareness, and accessibility to healthcare facilities, and to interpret how environmental and infrastructural conditions jointly influence rural health resilience.

By combining primary field data, household surveys, and spatial analysis, this study contributes to understanding rural environmental-health dynamics in a border-region context. It underscores the value of localized, data-driven epidemiological research for integrated rural health management and progress toward Sustainable Development Goals 3 and 6.

2. Study Area

The present study was conducted in the R.S. Pura sub-district of Jammu District, situated in the Union Territory of Jammu and Kashmir, India. The region lies in the southwestern part of Jammu District, adjacent to the international border, and represents a predominantly agrarian landscape interspersed with semi-urban settlements. Covering approximately 245 km², the sub-district comprises 193 villages with a total population of nearly 1.97 lakh (Census of India, 2011). The population density of 806 persons per km² reflects moderate settlement concentration, while agriculture and allied activities remain the main sources of livelihood.

To capture the local variability in environmental and healthcare conditions, two representative villages Suchetgarh and Kool Kalan were selected for detailed investigation. Both share similar socio-economic characteristics but differ in infrastructure and healthcare accessibility, offering an opportunity to compare the influence of environmental and institutional factors on health outcomes.

2.1 Suchetgarh Village

Suchetgarh is a medium-sized border village located about 31 km southwest of Jammu city and 8 km from the R.S. Pura headquarters (32°34'N, 74°40'E). Governed by a Gram Panchayat, it had a population of 946 across 208 households (Census of India, 2011). The village exhibits a distinctly rural character with limited healthcare infrastructure, low female literacy, and a substantial Scheduled Caste population (~45%). The sex ratio (866) and child sex ratio (614) fall below the Union Territory average, indicating gender disparity and social vulnerability (Census of India, 2011; NITI Aayog, 2022).

Environmental field observations revealed issues of open waste disposal, inadequate drainage, and limited awareness of sanitation practices, consistent with previous assessments of rural border settlements (Bose & Das, 2023; Sharma & Bhat, 2022). The area's dependence on agriculture and proximity to the border further constrain land-use flexibility and heighten exposure to environmental stressors. The spatial extent and location of Suchetgarh are shown in Figure 1.

[Figure 1 should appear here]

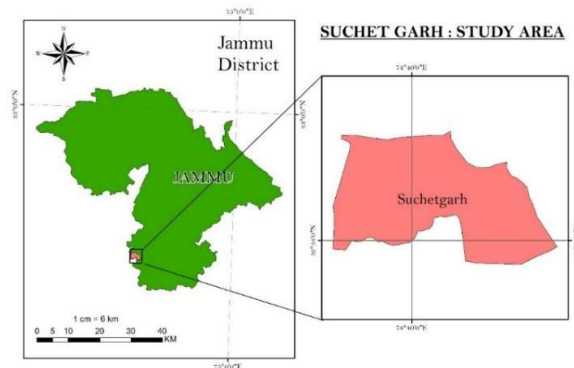


Figure 1. Location map of Suchetgarh Village.

Source: Field Survey and Digitized Map, 2023.

2.2 Kool Kalan Village

Kool Kalan, located in Bishnah tehsil of Jammu District, lies about 34 km from Jammu city (32.54°N , 74.79°E). It covers approximately 120 hectares and comprises 297 households. The settlement is spatially dispersed, and agriculture remains the backbone of the local economy. The area experiences a humid subtropical climate, with summer temperatures reaching up to 39°C and winter temperatures around 19°C . Most rainfall occurs between June and September, contributing to periodic waterlogging and seasonal health issues such as diarrhoea, fever, and vector-borne diseases. The geographical setting of Kool Kalan is depicted in Figure 2.

[Figure 2 should appear here]

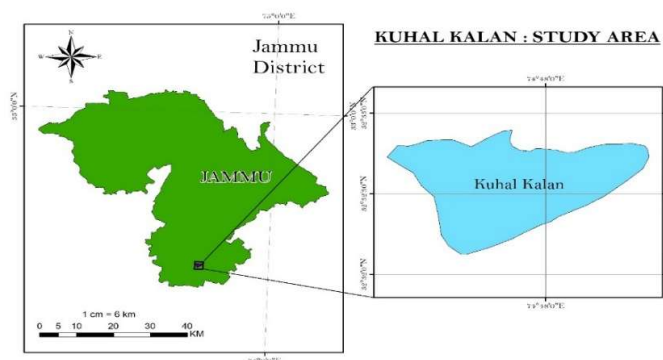


Figure 2. Location map of Kool Kalan Village.

Source: Field Survey and Digitized Map, 2023.

Both Suchetgarh and Kool Kalan exemplify the rural healthcare challenges of the Jammu region limited physical infrastructure, dependence on primary health centres, and low disease awareness. Their comparative analysis provides valuable insights into how environmental and infrastructural inequalities influence local health conditions and healthcare accessibility. The spatial relationship and route connectivity between the two villages are illustrated in Figure 3.

[Figure 3 should appear here]

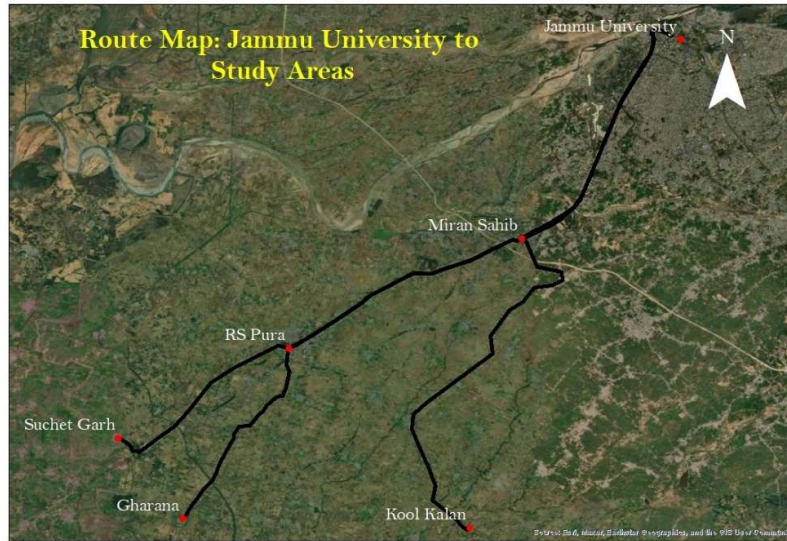


Figure 3. Route map showing connectivity between Suchetgarh and Kool Kalan.

Source: Author, 2023.

3. Literature Review

Rural India faces complex health challenges shaped by environmental, socio-economic, and infrastructural factors. Understanding how these determinants interact is essential for designing effective public health interventions. This review synthesizes contemporary studies up to 2023, focusing on environmental health, climate change, and epidemiological trends in rural settings, with attention to accessibility, awareness, and disease burden.

Environmental Health and Epidemiology in Rural India

Environmental determinants profoundly influence health outcomes in rural regions. Poor water, sanitation, and hygiene (WASH) conditions remain widespread, leading to elevated incidences of waterborne and vector-borne diseases. Nguyen et al. (2025) found that integrating clean water access with sanitation and hygiene education substantially reduces rural health risks

(*Frontiers in Public Health*). Inadequate healthcare infrastructure further exacerbates disease prevalence, as limited medical access delays treatment and increases morbidity rates.

Air pollution both ambient and household poses additional health threats. Adhikary et al. (2024) reported a strong association between PM_{2.5} exposure and daily mortality rates in India, with rural areas suffering disproportionately due to limited pollution control measures (*Wiley Online Library*). These environmental exposures strongly correlate with observed epidemiological patterns in developing regions.

Climate Change and Health Vulnerabilities

Climate change compounds existing health vulnerabilities in rural India. Extreme weather events such as heatwaves, erratic monsoon patterns, and floods are increasing in intensity. The Council on Energy, Environment and Water (2025) reported that nearly 57 % of Indian districts are at high to very high risk from extreme heat events, affecting rural communities disproportionately due to limited adaptive capacity (*Reuters*). Similarly, *The Lancet Countdown* (2024) highlights escalating health risks linked to climate change, including heat-related illness, food insecurity, and the expansion of vector-borne diseases like malaria and dengue. These findings emphasize the need to embed climate resilience within rural health policy frameworks.

Epidemiological Trends and Disease Burden

Epidemiological research indicates a dual burden of disease in rural India. While infectious diseases remain prevalent, non-communicable diseases (NCDs) are also rising due to lifestyle transitions and environmental exposures. Joshi et al. (2025) observed that rural India is increasingly witnessing NCD patterns once confined to urban populations (*Cureus*). Meanwhile, communicable diseases such as cholera and COVID-19 continue to persist, largely due to poor sanitation and healthcare infrastructure (WHO, 2023).

Access to Healthcare and Public Awareness

Healthcare access remains a critical determinant of health outcomes. Ganesan et al. (2023) reported that a large proportion of rural health centres in India lack essential WASH facilities, undermining service quality (*Lippincott Journals*). Public awareness also plays a vital role: communities with better health literacy exhibit improved preventive behaviour and treatment-seeking patterns. However, disparities in education and outreach continue to limit rural health outcomes.

Overall, literature underscores the intertwined nature of environmental, infrastructural, and behavioural factors in shaping rural health. Addressing these challenges demands a systems-based approach that integrates WASH improvements, healthcare strengthening, climate adaptation, and community participation. Future research should prioritise longitudinal monitoring and GIS-based assessments to better understand environmental health linkages in rural India.

4. Methodology

4.1 Study Design and Area

This study employed a cross-sectional and descriptive research design focusing on two representative villages Suchetgarh and Kool Kalan located in the southwestern part of Jammu District, India. Both areas are primarily agrarian and face recurring environmental and infrastructural challenges, including poor drainage, unsafe waste disposal, and seasonal waterlogging. The rationale for selecting these sites lies in their border proximity and ecological vulnerability, which together create distinctive environmental–health dynamics rarely addressed in previous research (Chakraborty & Singh, 2020; Gupta et al., 2022; Rao et al., 2023).

The research integrates environmental observation with epidemiological assessment, enabling a multidimensional understanding of rural health. This mixed environmental–epidemiological approach aligns with emerging frameworks in spatial health geography that advocate combining socio-environmental indicators for policy relevance and sustainable health planning (Joshi & Singh, 2021; Ganesan et al., 2023).

4.2 Data Collection and Survey Design

Primary data were collected through structured household surveys, direct field observations, and informal interviews. The questionnaire comprised five thematic sections:

1. **Demographic Profile:** age, gender, education, occupation, and income;
2. **Health Awareness and Disease History:** chronic and seasonal diseases, treatment behaviour, and self-reported health conditions;
3. **Environmental Conditions:** sanitation, waste management, drainage, water quality, and proximity to pollution sources;
4. **Healthcare Accessibility:** distance, quality, and satisfaction with medical services; and

5. Perception of Environmental Health Risks.

A total of 120 respondents (60 from each village) were selected through purposive random sampling to represent diverse age and occupational groups. Equal gender representation was maintained to capture differences in health awareness and accessibility patterns. The questionnaire was pre-tested among 10 respondents to ensure clarity and reliability, following which minor modifications were made.

Field surveys were supplemented by key informant interviews with local healthcare workers and Panchayat representatives, providing qualitative insights into institutional challenges, community participation, and policy implementation gaps. Observations on sanitation, drainage, and waste disposal were documented through geotagged photographs and field notes.

Respondents were informed about the academic purpose of the study and their participation was voluntary, with confidentiality maintained throughout.

4.3 Data Analysis and Interpretation

Data were coded and analysed using Microsoft Excel (2021) to derive frequency distributions, percentages, and comparative charts. Environmental and health variables were correlated descriptively to identify relationships between sanitation, disease prevalence, and healthcare accessibility.

Spatial analysis was conducted using ArcGIS 10.8 to generate thematic maps depicting settlement distribution, environmental risk zones, and healthcare locations. Spatial layers were prepared using Survey of India toposheets, GPS data, and open-access spatial datasets. GIS integration strengthened the visual representation of environmental–health linkages and demonstrated the potential of low-cost geospatial tools for rural epidemiological assessment (Ganesan et al., 2023; Joshi & Singh, 2021).

All figures were standardized with consistent symbology, scale, and colour transparency to ensure readability and interpretative clarity.

4.4 Validation and Reliability

To enhance data reliability, selected responses were cross-verified through repeat interviews with a subset of participants. Discrepancies were discussed directly with respondents for clarification. Triangulation was achieved by integrating multiple sources: survey data, government statistics, and GIS-based spatial datasets.

Secondary data from the Census of India (2011) and the District Statistical Handbook (2021) were used to contextualize socio-demographic indicators and infrastructural distribution. The combination of primary and secondary data improved analytical validity and strengthened the empirical foundation of the study (Nguyen et al., 2023; Ganesan et al., 2022).

4.5 Ethical Considerations

All participants were briefed about the study’s objectives and assured of anonymity. Verbal informed consent was obtained prior to participation, adhering to the ethical principles of the *Declaration of Helsinki* (World Medical Association, 2013). No personally identifiable information was collected, and all data were used solely for academic purposes.

5. Results and Discussion

5.1 Socio-Demographic Characteristics of Respondents

A total of 120 respondents were surveyed 60 each from Suchetgarh and Kool Kalan villages. Both communities exhibit predominantly agrarian economies, with nearly two-thirds (~65%) of participants engaged in agriculture or related activities. The remaining households depend on small-scale trade, wage labour, or public service. The gender distribution was almost balanced (52% male, 48% female), ensuring representation across households and age groups.

Educational attainment varied slightly between the two villages: Kool Kalan reported a higher proportion (~70%) of respondents with secondary or higher education compared to Suchetgarh (~60%). This difference likely stems from Kool Kalan’s proximity to Bishnah town, which provides better educational and healthcare access. Such disparities influence public health awareness and preventive behaviour, as observed in other rural districts of North India (Verma & Joshi, 2023; Singh & Mehra, 2023; Bose & Das, 2023). The socio-demographic characteristics of both villages are summarized in Table 1.

[Table 1 should appear here]

Particulars	Total	Male	Female
Total No. of Houses	208	-	-
Population	946	507	439
Child (0-6)	113	70	43
Schedule Caste	426	227	199
Schedule Tribe	0	0	0
Total Workers	453	289	164
Main Worker	421	0	0
Marginal Worker	32	17	15
Literate Population	634	376	258
Illiterate Population	312	131	181

Table 1. Socio-demographic profile of respondents in Suchetgarh and Kool Kalan
(Source: Field Survey, 2023)

Both villages reflect typical semi-rural border settlements marked by moderate literacy, limited occupational diversification, and dependency on natural resources. These socio-economic attributes shape both environmental exposure and health awareness levels, aligning with research linking socio-economic vulnerability to environmental health risks in rural India (Kumar et al., 2022; Chakraborty & Singh, 2020).

5.2 Environmental Conditions and Health Concerns

Environmental assessments revealed persistent sanitation and drainage issues across both villages. In Suchetgarh, 58% of households reported open dumping of solid waste in nearby drains or agricultural fields, while 47% in Kool Kalan practiced similar disposal methods. Monsoon-induced waterlogging was frequent in both areas, especially in low-lying settlements adjacent to paddy fields. Residents associated stagnant water with mosquito breeding and seasonal outbreaks of dengue and malaria.

These patterns mirror national trends where inadequate waste management and drainage systems in rural India contribute to recurring vector-borne and waterborne diseases (Nguyen et al., 2023; Sharma & Bhat, 2022). Additionally, 40% of respondents cited smoke from stubble burning as a cause of respiratory irritation, while 75% of farmers reported regular pesticide use without protective gear indicating low environmental health literacy. Comparable findings have been reported in Haryana and Punjab, where agrochemical exposure correlates with respiratory and dermatological ailments (Rana et al., 2021).

Air and water pollution were also emerging concerns. Poor ventilation, biomass fuel use, and lack of safe drinking water exacerbate the burden of respiratory and gastrointestinal diseases consistent with national epidemiological observations (WHO, 2023; Patel et al., 2022).

5.3 Prevalence of Chronic and Common Diseases

Field data indicate that diabetes, arthritis, and hypertension are the most prevalent chronic illnesses in both study areas. The distribution of chronic illnesses in the two villages is presented in Table 2.

[Table 2 should appear here]

CHRONIC DISEASES	VILLAGES	
	KOOL KALAN	SUCHET GARH
Diabetes	17%	24%
Arthritis	9%	10%
Thyroid	8%	8%
Lung Disease	6%	0%
Cervical	4%	0%
BP	3%	8%
Renal Calculi	3%	0%
Others	19%	10%
Nil	44%	40%

Table 2. Prevalence of chronic diseases in Suchetgarh and Kool Kalan

(Source: Field Survey, 2023)

A visual representation of these chronic disease patterns is shown in Figure 4.

[Figure 4 should appear here]

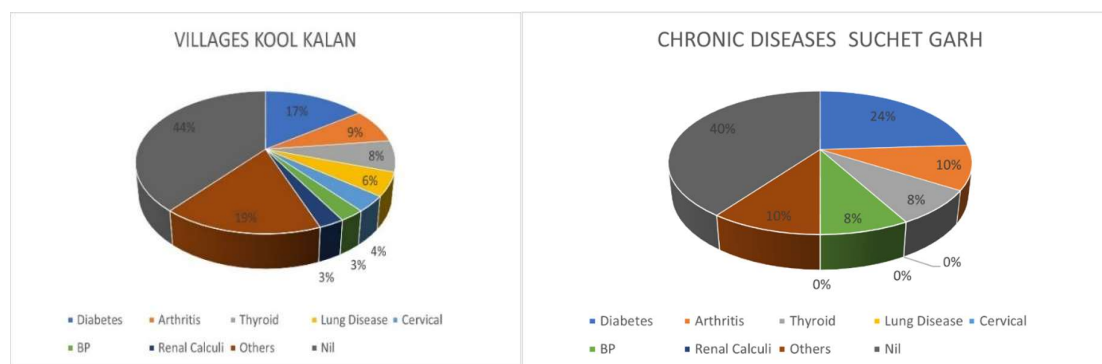


Figure 4. Pie chart showing chronic disease distribution in Suchetgarh and Kool Kalan

(Source: Field Survey, 2023)

In Suchetgarh, diabetes (24%), arthritis (10%), and hypertension (8%) were most common, while Kool Kalan reported slightly lower prevalence. Around 40–44% of respondents reported no chronic illness, suggesting that environmental rather than hereditary factors may play a more significant role in shaping local health outcomes. The commonly reported diseases among respondents are listed in Table 3.

[Table 3 should appear here]

COMMON DISEASES	VILLAGES	
	KOOL KALAN	SUCHET GARH
BP	24%	44%
Fever Cold	17%	18%
Body Pain	7%	0%
Allergies	7%	6%
Gastric\ Uric Acid	6%	10%
Diabetes	4%	0%
Arthritis	3%	0%

Urea	2%	0%
Headache	0%	14%
Cholesterol	2%	0%
Others	9%	0%
Nil	19%	8%

Table 3. Commonly reported diseases in Suchetgarh and Kool Kalan
(Source: Field Survey, 2023)

Figure 5 illustrates the distribution of these commonly reported illnesses.

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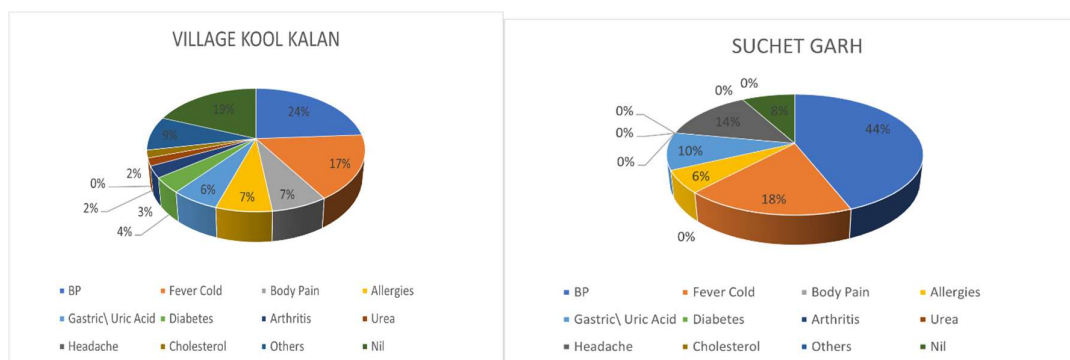


Figure 5. Pie chart showing common diseases in Suchetgarh and Kool Kalan
(Source: Field Survey, 2023)

Seasonal fluctuations were notable, with higher incidences of fever, cold, and gastric infections during monsoon months (July–September). These observations correspond with findings by Singh et al. (2023), who reported that humid, post-monsoon conditions significantly elevate vector- and waterborne disease risks in North India.

5.4 Accessibility and Quality of Healthcare Services

Healthcare accessibility emerged as a major determinant of local health outcomes. In Suchetgarh, the nearest Primary Health Centre (PHC) is located about 7–8 km away, compared to 4 km in Kool Kalan. However, respondents stressed that physical proximity did not necessarily translate into effective access. Irregular staff attendance, limited medicine availability, and lack of transport were major barriers. Respondents' rating of the functioning of the healthcare system is shown in Table 4.

[Table 4 should appear here]

S. No.	The way Health care system currently works?	Village Suchetgarh	Village Kool Kalan
1	Good	25	30
2	Fair	19	33
3	Poor	11	2

Table 4. Respondents’ rating of healthcare system performance
(Source: Field Survey, 2023)

Figure 6 presents a visual comparison of satisfaction levels with the existing healthcare facilities.

[Figure 6 should appear here]

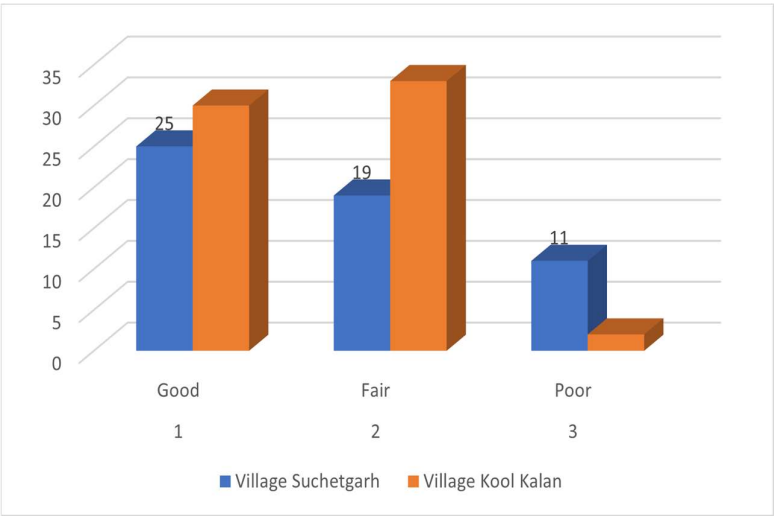


Figure 6. Bar chart showing satisfaction with healthcare functioning
(Source: Field Survey, 2023)

Only 45% of respondents in Suchetgarh and 58% in Kool Kalan rated healthcare access as “good” or “fair.” The remainder relied on private clinics or informal healers, particularly for minor ailments. The distribution of healthcare facilities visited by respondents is presented in Table 5.

[Table 5 should appear here]

VILLAGES	HCF VISITED		
	DISPENSARY	HOSPITAL	PRIVATE CLINIC
KOOL KALAN	64	19	17
SUCHET GARH	35	6	9

Table 5. Nearest healthcare facilities available to respondents
(Source: Field Survey, 2023)

A graphical comparison of facility usage patterns is shown in Figure 7.

[Figure 7 should appear here]

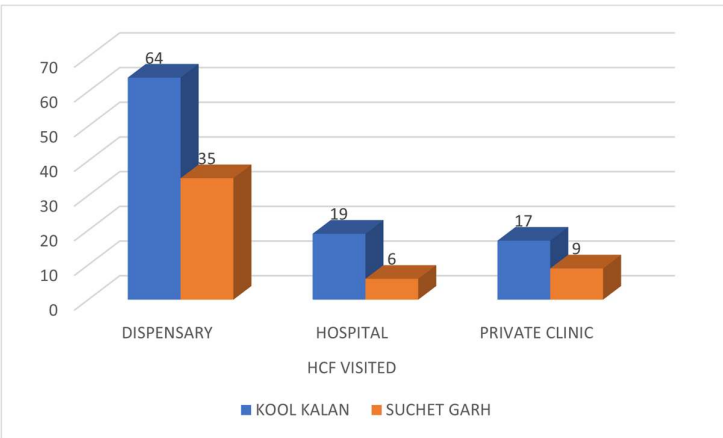


Figure 7. Bar chart showing distribution of healthcare facilities visited
(Source: Field Survey, 2023)

Spatial mapping (Figure 8) further highlights the uneven distribution of healthcare services within the R.S. Pura sub-district, where peripheral border villages such as Suchetgarh face longer travel times and delayed care.

[Figure 8 should appear here]

HEALTH CENTRES IN RANBIR SINGH PURA SUB-DISTRICT

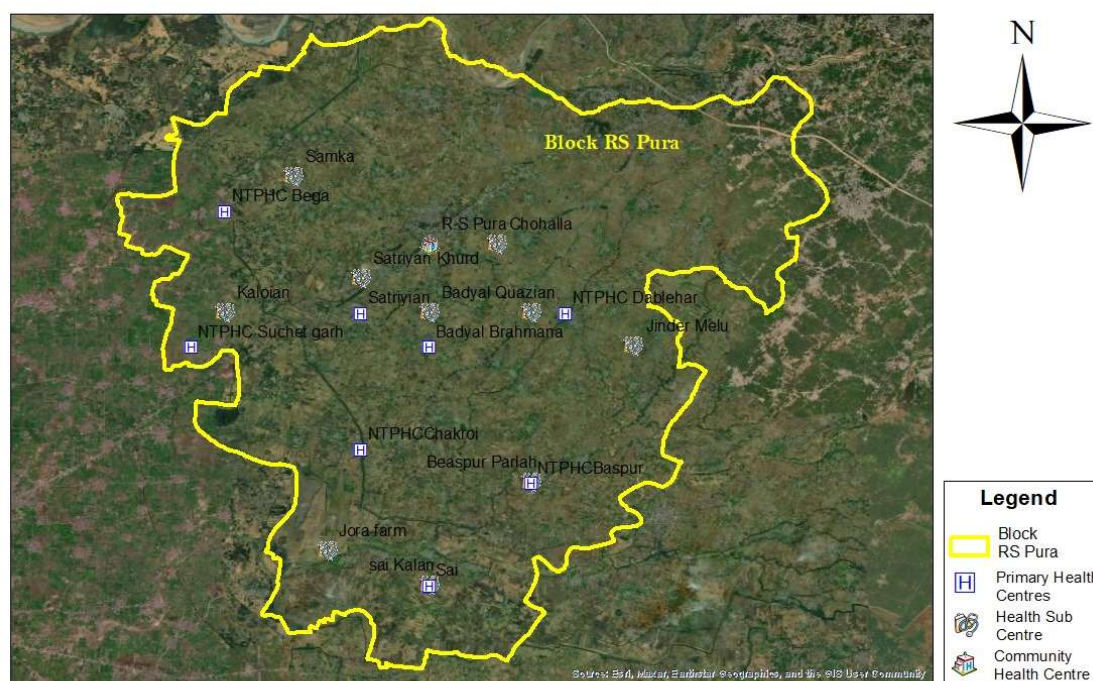


Figure 8. Spatial distribution of healthcare centres in R.S. Pura Sub-District

(Source: Author, ArcGIS 10.8, 2023)

These patterns are consistent with national findings where rural isolation and economic constraints limit healthcare utilization (Ganesan et al., 2022). Weak institutional outreach particularly regarding sanitation, hygiene, and preventive campaigns further widens the public health gap (Bose & Das, 2023).

5.5 Environmental–Epidemiological Linkages

Survey data reveal a clear correlation between environmental neglect, seasonal flooding, and disease prevalence. Around 70% of respondents attributed frequent illnesses during the monsoon to standing water and inadequate drainage. Female respondents and elderly residents were more likely to report respiratory or skin problems, largely due to indoor air pollution from biomass fuel use.

Such findings reinforce the argument that environmental determinants serve as fundamental epidemiological drivers in rural India. Narang et al. (2022) and WHO (2023) confirm that localized environmental conditions such as waste accumulation, poor ventilation, and contaminated water are statistically linked with higher infection and morbidity rates.

Although 80% of households reported access to toilets, only 60% maintained regular cleanliness, primarily due to intermittent water supply. Open drains and livestock proximity further increased microbial exposure risks. The observed disease spectrum particularly diarrhoea, typhoid, and vector-borne infections aligns with environment-linked epidemiological conditions widely documented in developing countries (Nguyen et al., 2023).

5.6 Awareness, Perception, and Community Practices

Health and environmental awareness varied between the two villages. In Kool Kalan, 65% of respondents acknowledged the importance of waste segregation and mosquito prevention, compared to 50% in Suchetgarh. Despite moderate awareness, behavioural adoption remained low due to lack of infrastructure and entrenched habits, as observed in similar rural settings across India (Kumar & Das, 2023).

Participation in community health initiatives such as Swachh Bharat Abhiyan was minimal, with only 18% reporting any involvement, reflecting post-pandemic behavioural gaps in sanitation governance (Das & Mukherjee, 2022). Public health campaigns were limited mainly to vaccination drives, with few programs addressing sanitation or waste management.

However, awareness levels improved marginally after the COVID-19 pandemic, particularly among households near PHCs, echoing post-pandemic behavioural changes documented in recent studies (WHO, 2022; Singh & Mehra, 2023).

5.7 Summary of Findings

The analysis demonstrates that health disparities in Suchetgarh and Kool Kalan are driven by a combination of environmental neglect, infrastructural inadequacy, and socio-economic vulnerability. While Kool Kalan performs slightly better in healthcare access and awareness, both villages experience recurrent, preventable diseases stemming from poor sanitation and limited health literacy.

These findings reaffirm the need for an integrated environmental–epidemiological framework in rural planning. Strengthening primary healthcare networks, ensuring sustainable sanitation, and embedding community-based awareness programs can significantly reduce the disease burden. The results support global calls by WHO (2023) and UNDP (2023) for localized, data-driven strategies to achieve Sustainable Development Goals (SDG 3: Good Health and Well-being, and SDG 6: Clean Water and Sanitation).

6. Conclusion

This study explored the interrelationship between environmental conditions, healthcare accessibility, and public health awareness in two semi-rural border villages Suchetgarh and Kool Kalan of the R.S. Pura sub-district, Jammu, India. Despite moderate literacy levels and relative proximity to urban centres, both settlements continue to face chronic environmental and infrastructural challenges that directly influence disease prevalence and community well-being.

The findings reveal that poor drainage, open waste disposal, and periodic waterlogging during monsoon months are the primary environmental stressors contributing to recurrent vector-borne and waterborne illnesses. Limited healthcare infrastructure, irregular service availability, and economic constraints further exacerbate health vulnerability. While Kool Kalan demonstrates slightly better healthcare access and awareness levels than Suchetgarh, both villages exhibit comparable patterns of environmentally linked ailments such as fever, diarrhoea, respiratory problems, and chronic diseases like diabetes and hypertension.

These results reinforce the importance of adopting an integrated *environmental–epidemiological approach* to rural health planning. Environmental determinants are not merely background factors but active contributors to health inequities, particularly in border and agrarian regions. This study thus provides empirical, field-based evidence supporting the inclusion of environmental management within public health frameworks.

6.1 Policy Implications and Recommendations

The findings have several practical implications for sustainable rural health improvement:

1. Strengthen Primary Healthcare Networks:

Establish satellite healthcare units or mobile medical camps in peripheral villages such as Suchetgarh to reduce travel time and improve disease surveillance.

Strengthening coordination between local Panchayats and district health departments can enhance service continuity.

2. Improve Water, Sanitation, and Waste Infrastructure:

Implement decentralized waste management systems and construct covered drainage channels to mitigate monsoon-related stagnation. Promoting household-level sanitation maintenance and safe pesticide handling can reduce exposure risks.

3. Integrate Environmental Management with Health Planning:

Environmental health assessments should become part of routine rural health monitoring. Integrating WASH indicators and local environmental audits into PHC operations can provide early warnings for potential disease outbreaks (Nguyen et al., 2023; WHO, 2023).

4. Community Awareness and Behavioural Change:

Health literacy programs, particularly targeting women and farmers, can encourage preventive practices such as waste segregation, mosquito control, and safe water storage. Schools and self-help groups can play a crucial role in community-based awareness dissemination.

5. Adopt the “One Health” Framework:

Recognizing the interconnection between human, animal, and environmental health is vital in agrarian regions. Promoting cross-sector collaboration among healthcare workers, veterinarians, and environmental agencies can support early detection and control of zoonotic and vector-borne diseases.

6. Encourage Localized Research and Data Systems:

Village-level data collection on water quality, disease patterns, and healthcare access should be institutionalized. Encouraging local universities and NGOs to collaborate on environmental health mapping will build capacity for evidence-based policymaking.

6.2 Broader Relevance

The study contributes to the limited body of literature on environmental epidemiology in border-region contexts of South Asia. Its results highlight how micro-scale environmental risks, when combined with infrastructural neglect, can translate into broader public health disparities. The case of Suchetgarh and Kool Kalan illustrates that improving environmental conditions is not only a matter of infrastructure but also of governance, awareness, and equity.

By bridging environmental science and public health, this research aligns with the *Sustainable Development Goals (SDG 3 – Good Health and Well-being, and SDG 6 – Clean Water and Sanitation)*. It advocates for integrated planning that addresses both ecological and social

determinants of health an approach increasingly recognized as critical to achieving sustainable development in rural and peri-urban India (UNDP, 2023; Patel et al., 2022).

DECLARATIONS

Ethical approval

Formal institutional ethical approval was not required for this study as it involved a non-invasive, observational, questionnaire-based household survey conducted for academic research purposes. The study did not include any clinical interventions, biological sampling, or collection of personally identifiable or sensitive personal data and posed minimal risk to participants.

Consent to participate

Informed verbal consent was obtained from all participants prior to data collection. The survey was conducted through door-to-door household visits, during which participants were clearly informed about the purpose and scope of the study, the voluntary nature of participation, and their right to decline or withdraw at any time. Verbal consent was considered appropriate given the non-invasive nature of the study and the community-based rural setting.

Consent to publish

All participants consented to the use of anonymized data for academic research and publication purposes. No identifying information is included in the manuscript.

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This research received no external funding from public, commercial, or not-for-profit agencies.

Conflicts of Interest

The author declares no conflict of interest.

Data Availability

The data supporting the findings of this study are available from the corresponding author upon reasonable request.

Author Contributions

The author conceptualized the study, conducted the field survey, analyzed the data, prepared maps and figures, interpreted the results, and wrote the manuscript.

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